



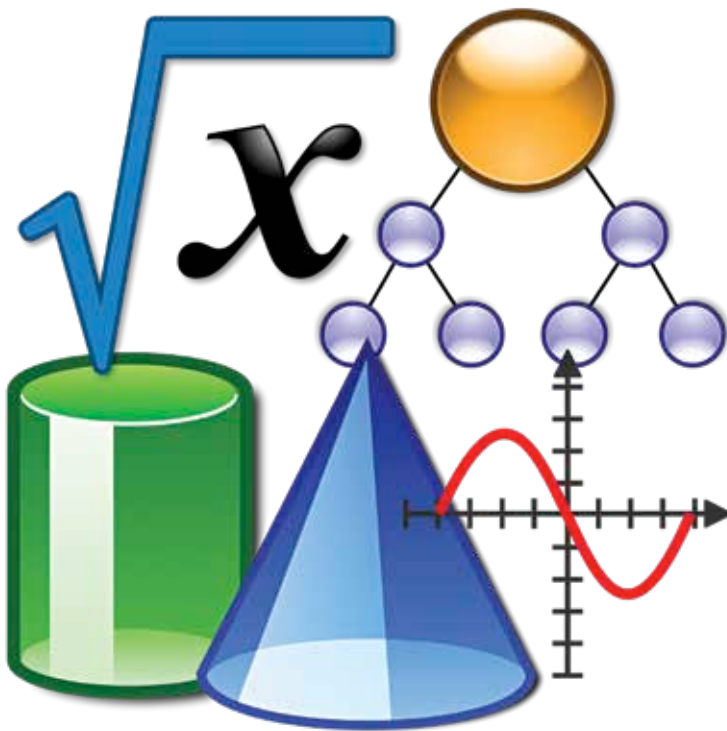
No sound in SpaceX

SpaceX employees who issued an open letter against Elon Musk have been fired.
<https://digit.in/jul22-52>



Array of Hope

The United States army deploys its first floating solar array.
<https://digit.in/jul22-53>



GREATEST EQUATIONS

More impactful than you think

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he world of mathematics has a lot of equations that are used to perform complex calculations.

However, there are a few equations that mathematicians and scientists alike can confidently say have had a significant impact on research, our understanding of our universe, and the development of technology. In this story, we have singled out 10 equations which are considered responsible for some of the greatest discoveries in the history of science and technology.

THEORY OF RELATIVITY

The Theory of Relativity, put forth by Albert Einstein, is arguably one of the greatest findings in the field of science. This equation allows researchers and scientists to look at the concept of reality through a completely different lens. There can be no list of special equations without the inclusion of this one. This equation states that the energy dispersed by an object moving at the speed of light is the product of the squared value of the speed of light and the object's

$$E=mc^2$$

mass. It played an instrumental role in atomic energy development and also refined several daily use concepts like GPS-based navigation.

CALCULUS

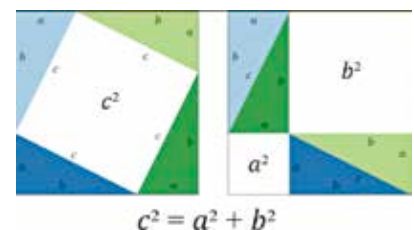
The nightmare of every math and physics student, Calculus holds great significance in scientific and technological advancement. First conceptualised in ancient times, calculus was once called "the calculus of infinitesimals". These equations essentially define continuous change. They are

$$f(x) = \int_0^x f'(t) dt$$

divided into two different branches – integral calculus and differential calculus. Calculus was famously used by Isaac Newton in the modern era of research and development to describe planetary motion around the sun.

PYTHAGORAS THEOREM

Another blast from the past for math students, the Pythagoras Theorem and the equation related to it, is of great importance in the world of R&D. It can easily be regarded to be one of the founding pillars of all geometry. While there may still be some conflict regarding the attribu-



tion of this concept's discovery, its impact on mathematical calculations (especially those related to geometry) cannot be negated. Both geometrical and algebraic proofs of this equation have been seen over the years, the earliest being thousands of years old.

COMPLEX NUMBERS

Complex Numbers, when they were